

Uzair Khan

Network Operations Engineer · Manassas, VA
CompTIA CySA+ (CS0-003, valid through 2027) | English
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SUMMARY

- IT Support & Network Operations Engineer with 8+ years supporting enterprise, government, and global environments in high-availability, SLA-driven settings spanning satellite operations, correctional facility WAN, and state government IT.
- **Networking:** OSPF, BGP, EIGRP, MPLS, Layer 2/3 switching (VLAN, STP, LACP), DNS, DHCP, NTP, IPAM, VPN, load balancers, LAN/WAN/MAN design and administration.
- **Endpoint & Systems:** Microsoft Intune, Azure AD/Entra ID, SCCM, Autopilot, Active Directory; enterprise imaging, policy enforcement, conditional access, and device lifecycle management.
- **Monitoring & Telemetry:** SolarWinds NPM/NTA, Splunk SIEM, SNMP, NetFlow, syslog; incident triage, root-cause analysis, and SLA-driven escalation leadership.
- Satellite platforms: iDirect, Hughes, Comtech. DVB-S2 carrier management, ModCod optimization, SCPC/MCPC configuration. NOC operations across maritime, aero, and fixed VSAT environments.
- Experienced in change management, SOP/KB authoring, Jira/Confluence administration, PowerShell/Python automation, and infrastructure documentation.

PROFESSIONAL EXPERIENCE

IT Functional Analyst II / Systems Administrator

2024–2025

MDThink – DoIT: Maryland Benefits (SNAP/DHS) · Linthicum, MD

- **Endpoint Fleet Management:** Managed enterprise endpoint lifecycle using Microsoft Intune, SCCM, and Autopilot for 200+ Windows 10/11 devices (laptops/tablets) supporting onsite and remote software engineers on SNAP benefits systems for Maryland DHS. Deployed, imaged, and configured devices with role-based access, VPN credentials, and application packages for day-one onboarding.
- **Azure AD / Entra ID:** Administered Azure AD Conditional Access policies, group-based licensing, and MFA enforcement via Entra ID. Managed user provisioning, de-provisioning, and access reviews; served as ticket owner for Jira-based credential and IAM access requests coordinated across system teams.
- **Policy Deployment & Automated Testing:** Authored PowerShell automation scripts deployed via SolarWinds to test configuration policy push success across hundreds of endpoints. Validated GPO and Intune baseline propagation; flagged failures and remediated in coordination with systems architect following change control procedures.
- **Firmware & Patch Management:** Executed firmware updates and patch deployments through endpoint management console per systems architect directives; evaluated change risk criteria, held updates when thresholds not met, and maintained audit trails per change management protocol.
- **Asset Inventory Automation:** Built Python/PowerShell scan script to inventory state-owned and vendor laptops and electronics; integrated output with custom Jira inventory schema, automated report generation to Confluence, and distributed digest reports to VP, team lead, and manager.
- **Jira Dashboard Engineering:** Designed multi-panel Jira dashboards tracking ticket queues, shipping status, storage inventory, and device lifecycle using custom JQL queries, pie chart gadgets, and Confluence KB widget integrations. Built dedicated dashboards for team leads and management.
- **Knowledge Base & SOP Authoring:** Authored enterprise KB articles covering endpoint onboarding/offboarding procedures, Wi-Fi failover recovery, FedEx/USPS device intake, shared mailbox setup, vendor communication standards, and PTO calendar processes. Documented end-to-end workflows adopted organization-wide.
- **Conversational Agent (AI):** Prototyped AI-powered IT helpdesk chatbot for Google Spaces to handle policy Q&A (e.g., "Can I install this software?") reducing tier-1 ticket volume and enabling self-service for software engineers.
- **Physical Infrastructure:** Conducted on-site infrastructure audits: verified wall jack connectivity, traced and matched switch port assignments, managed copper patch panel labeling, replaced HDMI/power cabling, and maintained network closet organization. Provided on-site AV and network support for executive leadership presentations.

- **ServiceNow ITSM:** Managed incident, request, and change tickets in ServiceNow; maintained SLA compliance, escalated P1/P2 incidents per runbook, and produced weekly operational status reports.

Impact: Improved endpoint deployment velocity and reduced provisioning errors through automation. Inventory script eliminated manual asset tracking for 200+ state-owned devices. Self-service KB adoption reduced repeat helpdesk tickets.

Tools/Skills: Microsoft Intune, SCCM, Autopilot, Azure AD/Entra ID, Active Directory, GPO, PowerShell, Python, SolarWinds, ServiceNow, Jira (JQL, dashboards), Confluence, Google Workspace, Windows 10/11, change management, SOP/KB authoring

Network Engineer – Network Operations Center

2022–2024

SES Networks · Manassas, VA

- **Global Satellite NOC (24x7):** Tier 2/3 NOC operations for a global satellite and IP network serving maritime vessels, fixed VSAT sites, and aero (inflight connectivity) customers. Monitored and managed iDirect, Hughes, and Comtech modem platforms across HTS and FSS orbital assets. Supported SKALA customers and Virtual Network Operators (VNOs), serving as primary NOC point of contact for their service events.
- **P1dB Saturation Testing:** Coordinated and bridged P1dB (1 dB compression point) transmit power tests for SKALA and VNO customers. Acted as call bridge between customer, ground station RF engineers, and NOC; managed test execution, monitored uplink power levels, documented results, and cleared the carrier for operational use post-test.
- **Site Commissioning & Decommissioning:** Commissioned new fixed terminals and small vessel VSAT sites end-to-end: verified azimuth/elevation alignment, confirmed modem registration, validated link margins, and signed off on service activation. Decommissioned legacy sites and updated NMS inventory accordingly.
- **Beam Switching & Vessel Management:** Executed manual beam switches across all operational regions to optimize satellite coverage, load-balance capacity, and migrate vessels to better-performing orbital slots. Managed bandwidth package updates per customer contract terms; coordinated with capacity planning on regional rebalancing.
- **DVB-S2 / RF Management:** Performed real-time ModCod (Modulation and Coding Scheme) adjustments on DVB-S2 forward and return carriers to optimize throughput under variable atmospheric conditions. Managed SCPC and MCPC carrier configurations and transponder utilization across active beams.
- **OSPF & BGP Operations:** Administered OSPF on the backbone network; executed CLI commands to check neighbor adjacencies, review routing flags, and verify route state (up/down). Performed BGP diagnostics on vessel CPE: checked neighbor state, reviewed prefix advertisements, analyzed logs for session resets and path selection issues.
- **iDirect Modem Management:** Pulled and analyzed iDirect modem logs to diagnose connectivity failures, registration issues, and signal quality degradation. Checked Ec/No, symbol rate, and upstream/downstream stats to isolate RF vs. network layer faults.
- **Backbone Network Monitoring:** Monitored terrestrial backbone links during outage events; checked interface status, reviewed routing convergence, and escalated to backbone engineering with structured fault summaries when terrestrial path failures impacted satellite traffic.
- **Change Request Management:** Authored and issued CR (Change Request) notifications to end customers for planned maintenance windows; tracked CR lifecycle in ServiceNow from creation through resolution and customer confirmation.
- **Network Monitoring & Telemetry:** Leveraged SNMP polling, NetFlow analysis, and syslog aggregation via multi-vendor NMS platforms to detect performance degradation, packet loss, and latency anomalies. Correlated alerts across RF, routing, and hardware layers to isolate root cause.
- **Aero Operations:** Monitored live inflight routes in real-time via airline ops dashboards; managed bandwidth allocation, beam handoffs, and ModCod transitions during active flights to maintain SLA throughput commitments for cabin IFC (inflight connectivity).
- **Splunk SIEM:** Onboarded new ground station and network node log sources into Splunk SIEM; built SPL searches and threshold alerts for anomaly detection, reducing MTTD for network faults across 30+ nodes.
- **Incident Command:** Led incident bridges during service-affecting events; coordinated resolution across satellite ops, terrestrial engineering, ground station teams, and vendor NOCs. Authored RCA reports and maintained shift handover documentation in ServiceNow and Jira.
- Mentored junior NOC analysts on satellite link troubleshooting, protocol-level diagnostics, P1dB test procedures, and escalation runbooks.

Impact: Maintained 99.9%+ availability across maritime and aero customer links. P1dB test bridge coordination reduced SKALA customer activation delays. Splunk integration improved fault correlation across 30+ nodes.

Tools/Skills: iDirect, Hughes, Comtech, DVB-S2, SCPC/MCPC, ModCod, P1dB testing, BGP, OSPF, MPLS, Cisco IOS/NX-OS, Juniper JunOS, SNMP, NetFlow, syslog, Wireshark, Splunk SIEM, NMS (multi-vendor), ServiceNow, Jira, CR management

Enterprise Network Technician – Network Operations Center

2021–2022

ViaPath · Falls Church, VA

- **WAN/LAN Monitoring:** Monitored network infrastructure across 100+ correctional facility sites using SolarWinds NPM/NTA; responded to SNMP-triggered alerts within 15-minute SLA windows, performed log analysis, and executed root-cause diagnostics for outages and performance degradation.
- **Incident Triage & Carrier Coordination:** Triageed events using IR playbooks; coordinated with on-site contacts to verify physical layer conditions (power, hardware status) and with Local Exchange Carriers (LECs) to validate maintenance windows and circuit-level disruptions.
- **Change Management:** Tracked planned maintenance through structured Change Request (CR) processes; cross-referenced schedules with live alert queues to suppress false-positive escalations and maintain accurate operational status across sites.
- **Satellite Failover:** Initiated ViaSat satellite failover circuits for remote correctional facilities without terrestrial fiber; documented activation and restoration procedures per site runbook to ensure continuity of correctional management systems.
- Provisioned new facility sites and decommissioned legacy locations; dispatched field technicians with hardware replacement SOPs. Provided junior analyst mentorship on escalation and diagnostic best practices.

Impact: Improved uptime across mission-critical correctional facility infrastructure through rapid SLA response, structured carrier coordination, and redundancy documentation.

Tools/Skills: SolarWinds NPM/NTA, SNMP, WAN/LAN monitoring, Change Request (CR) processes, ViaSat failover, LEC coordination, ServiceNow, hardware diagnostics, runbook authoring

Enterprise Account & Operations Support

2017–2020

Carfax · Centreville, VA

- **API & POS Integration Support:** Tier 2 support for CARFAX RESTful API and POS software integrations at automotive service providers; diagnosed data transmission failures, field mapping errors, and OAuth/authentication issues between dealer management systems and CARFAX vehicle history APIs.
- **Remote Deployment & Configuration:** Remotely deployed and configured CARFAX integration software on client POS systems via remote desktop; validated end-to-end data pipeline from point-of-sale capture through vehicle history report generation.
- Managed ticket lifecycle in Salesforce CRM; documented root cause, resolution steps, and KB articles for recurring integration failures. Collaborated with engineering to escalate upstream API defects with structured reproduction steps and log artifacts.

Impact: Reduced integration failure rate and support backlog through proactive documentation and structured escalation workflows.

Tools/Skills: RESTful APIs, POS systems, Salesforce CRM, remote desktop tooling, data integration troubleshooting, technical documentation

Retail Operations Manager

2012–2016

AT&T Mobility · Oakton, VA

- **OPUS NCR Tablet POS:** Operated AT&T's OPUS NCR tablet-based point-of-sale system — one of the first tablet POS deployments in telecom retail. Managed end-to-end transactions, inventory, and system error resolution within the OPUS interface. Trained and led team of 10 on OPUS workflows.
- **Backend CLI Provisioning:** Used AT&T's backend command-line provisioning system to activate wireless service, push SIM-level configurations, assign rate plans, and bring new lines of service online. Diagnosed and resolved provisioning failures at the carrier backend level, including feature code errors and order management exceptions.
- **DSAT / DIRECTV Backend:** Configured and activated DIRECTV satellite installations via DSAT backend; managed account setup, equipment provisioning, and technician dispatch coordination for residential and commercial bundled service activations.
- **MDM & Device Activation:** Provisioned smartphones, tablets, and IoT devices; performed MDM enrollment for business accounts and configured enterprise email profiles, VPN settings, and mobile app management (MAM) policies for commercial customers.
- Owned store-level revenue targets across wireless, DIRECTV, security systems, and business software (geo-fencing, lead generation). Executed workforce planning, compliance audits, and product launch campaigns with district leadership.

Impact: Operated one of AT&T's earliest OPUS tablet POS environments. Backend CLI proficiency reduced provisioning errors and customer wait times. Consistent revenue growth across consumer and commercial technology lines.

Tools/Skills: OPUS NCR tablet POS, AT&T backend provisioning CLI, DSAT/DIRECTV systems, MDM device activation, CRM (Siebel/Salesforce), KPI dashboards, workforce management

TECHNICAL SKILLS

Networking: OSPF, BGP, EIGRP, MPLS, VLANs, STP, LACP, DNS, DHCP, NTP, IPAM, VPN, load balancers, LAN/WAN/MAN

Satellite / RF: iDirect, Hughes, Comtech, DVB-S2, SCPC/MCPC, ModCod, P1dB testing, beam switching, VSAT commissioning, HTS/FSS, aero IFC, maritime NOC, SKALA/VNO support

Monitoring: SolarWinds NPM/NTA, Splunk SIEM, SNMP, NetFlow, syslog, Wireshark, tcpdump, NMS (multi-vendor)

Endpoint / MDM: Microsoft Intune, SCCM, Autopilot, Azure AD/Entra ID, Active Directory, GPO, Windows 10/11

Automation: PowerShell, Python — policy testing, asset inventory, web scraping, LLM/RAG pipeline development

ITSM / Collab: ServiceNow, Jira (JQL, dashboards), Confluence, Google Workspace, Zoom, Microsoft Teams, WebEx

Documentation: SOP/KB authoring, CR notification management, RCA reports, runbook development, technical writing

Infrastructure: Copper/fiber cabling, patch panel management, switch port mapping, rack/elevation drawings

PROJECTS

TankPit Discord Bot & RAG AI System

Personal Project · github.com/uk0430

- **Architecture:** Built a full-stack Discord bot with a hybrid RAG + SQL AI Q&A engine. Deployed local Mistral 7B LLM and nomic-embed via llama.cpp; embedded 13,250 player profiles into a vector store for cosine similarity retrieval. Integrated SQLite for structured queries.
- **Data Pipeline:** Designed and ran async web scrapers (Python/aiohttp/Playwright) to collect 13,250+ player profiles and 5,000+ days of community bulletin board data. Built rebuild_bible.py to chunk, embed, and version the knowledge base.
- **Stack:** Python, discord.py, SQLite, llama.cpp, nomic-embed, Mistral 7B, PIL (image generation), asyncio, PowerShell automation, WSL2/Ubuntu.

Home Network Monitoring Lab (Raspberry Pi)

Personal Project · In Progress

- **Overview:** Deployed a Raspberry Pi-based network monitoring solution on home LAN; configured SNMP polling, traffic capture, and alerting to mirror enterprise NOC workflows in a home lab environment. Exploring integration with SolarWinds-style dashboards.

EDUCATION

Northern Virginia Community College

Associate Degree in Business Administration (AA/AS)

Strayer University

B.S. in Information Technology, In Progress — Prerequisite coursework at NVCC (Starting March 2026)

CERTIFICATIONS

- CompTIA CySA+ (CS0-003, valid through 2027)